

Polynomial, MLP and AR(1) Closures for the Fast Variables of Two-Scale Lorenz-96: Forecast Skill, Stability and Climate

Agate-71N

Stemma seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/lorenz96-learned-parametrisation>

Abstract—We use the two-scale Lorenz-96 system as a small testbed for learned subgrid parametrisation and replace its 256 fast variables by three closures of the slow state: a regression polynomial, a small MLP, and the polynomial plus AR(1) red noise. All three are reimplemented, trained offline on the same data, and scored against the full two-scale truth on forecast valid time, free-run stability and the mean, variance, PDF and spatial spectrum of the slow variables, over five seeds and two scale-separation regimes ($c = 10$ and $c = 4$). All closures raise valid time from 0.363 (no closure) to 1.252–1.421 model time units at $c = 10$ and none diverged in any run. The deterministic closures forecast best, and the registered tests do not distinguish the MLP from the polynomial when both see only the local slow variable. The AR(1) closure has the best climate at $c = 10$ (spectral error 0.056 versus 0.144 for the polynomial) but a shorter valid time (1.252 versus 1.398); at $c = 4$ all learned closures sit near the sampling-noise floor of the climate metrics. In three-seed ablations, giving the MLP neighbouring slow variables lowered its mean climate error (and, with seven inputs, raised valid time), the AR(1) closure no longer had the lower spectral and PDF error when deployed at forcings $F=18$ and $F=22$ (possibly because its noise amplitude was fitted at $F=20$; not isolated by a run), and a degree-2 polynomial gave a worse climate than a linear one despite a higher offline R^2 . This is a small CPU study of an idealised toy system with a single 8-variable configuration and offline training only.

Index Terms—Lorenz-96, subgrid parametrisation, stochastic closure, climate emulation, machine learning

I. Introduction

Weather and climate models cannot resolve every process, so unresolved “subgrid” effects are replaced by parametrisations. Data-driven parametrisations learn this map from high-resolution simulation: Rasp et al. trained a deep network on a multiscale model and obtained stable multi-year prognostic runs that reproduced the mean climate and aspects of variability [1]. Whether a closure that is accurate offline is also good when coupled to the resolved model is a separate question. Rasp noted that online simulations with offline-trained machine-learning parametrisations in Earth system models were frequently plagued by

instabilities and biases, proposed coupled online learning as a remedy, and illustrated the concept in the Lorenz-96 model [2]; Frezat et al. train subgrid parametrisations online for non-differentiable solvers by means of a neural emulator [3]. Stochastic closures are the other classical approach: the two-scale Lorenz-96 system [4] has been used to study the effects of stochastic parametrisations [5] and their link to model uncertainty [6], and learned stochastic variants use GANs [7] or recurrent networks with red noise [8].

These studies differ in their system settings, input features and scores, so it is hard to read off from them how the three most basic closures compare. We use the two-scale Lorenz-96 system as a cheap, idealised testbed and ask: with identical training data, inputs and metrics, how do a regression polynomial, a small MLP and a polynomial with AR(1) noise differ in (i) short-term forecast skill, (ii) long-rollout stability and (iii) the climate of the slow variables, and does the ranking change with scale separation and a forcing shift? This is a controlled small-scale comparison, not a new method. Our contributions are: a fully specified protocol with a sampling-noise floor from an independent truth run; five-seed results at two scale-separation regimes with the hypotheses and tests fixed before the runs; and ablations on polynomial degree, noise amplitude and memory, MLP input width and forcing shift, including cases where a closure fails.

II. Related Work

Stochastic parametrisation in Lorenz-96. The two-scale system was introduced by Lorenz [4]. Wilks examined the effects of stochastic parametrisations in it [5], and Arnold et al. studied stochastic parametrisations and model uncertainty there [6]; our AR(1) closure follows this line, with the mean closure fitted by regression and the noise fitted to its residual. Outside this toy system, Shutts proposed a kinetic-energy backscatter algorithm for use in ensemble prediction systems [9].

Learned deterministic closures and coupled behaviour. Rasp et al. learned all subgrid processes of a multiscale atmospheric model with a deep network [1]. Rasp observed that online simulations with offline-trained machine-

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learning parametrisations in Earth system models were frequently plagued by instabilities and biases, and proposed coupled online learning, a second training stage run alongside a high-resolution simulation, illustrating it in the Lorenz-96 model [2]; a further approach trains parametrisations online for non-differentiable solvers with a neural emulator [3]. We only train offline and measure what happens when the closure is coupled, so our stability result is limited to that setting.

Learned stochastic closures. Gagne et al. used generative adversarial networks as stochastic parametrisations in Lorenz-96 [7]; Parthipan et al. combined a recurrent network with red noise in a probabilistic framework [8]. These are richer than our AR(1) closure and are out of scope here.

Data-driven emulation of Lorenz-96. Chattopadhyay et al. compared echo-state, feed-forward and LSTM networks for short-term prediction and long-term statistics of a multiscale Lorenz-96 system [10], and Brajard et al. combined data assimilation with a neural network to emulate a dynamical model from sparse noisy observations, with the 40-variable Lorenz-96 model as the test case [11]. Those works emulate the system itself; we instead keep the slow-variable equations and learn only the subgrid term.

III. Method

Truth. The two-scale Lorenz-96 system [4] has here $K=8$ slow variables X_k and $J=32$ fast variables $Y_{j,k}$ per slow one (Y on a ring of KJ points):

$$\dot{X}_k = -X_{k-1}(X_{k-2} - X_{k+1}) - X_k + F - U_k, \quad (1)$$

$$U_k = \frac{hc}{b} \sum_j Y_{j,k}, \quad (2)$$

$$\dot{Y}_i = -cb Y_{i+1}(Y_{i+2} - Y_{i-1}) - c Y_i + \frac{hc}{b} X_{k(i)}, \quad (3)$$

with $h=1$, $b=10$, $F=20$ and $c \in \{10, 4\}$ (time-scale ratio). The subgrid tendency $U_k(t)$ is recorded every $\Delta = 0.005$ time units.

Closed slow model. Every closure replaces U_k by $\hat{U}_k = g(X_k \dots) + e_k$ in $\dot{X}_k = f_k(X) - \hat{U}_k$ with $f_k = -X_{k-1}(X_{k-2} - X_{k+1}) - X_k + F$, integrated with RK4 at step Δ ; g is evaluated at every RK4 stage, e_k is held fixed within a step. The same g is shared by all sectors k .

No closure. $\hat{U}_k = 0$ (lower reference).

Polynomial (reimplemented). $g(X_k) = \sum_{p=0}^n a_p z^p$, $z = (X_k - \mu)/s$, with a from least squares, $\min_a \sum_{t,k} (U_k(t) - g(X_k(t)))^2$; degree $n=4$ fixed a priori.

MLP (reimplemented). $g(x) = W_3 \tanh(W_2 \tanh(W_1 x + b_1) + b_2) + b_3$ with hidden width 32, input $x = (X_{k-s}, \dots, X_{k+s})$ with $s=0$ (local, same input as the polynomial) in the main comparison; standardised inputs and targets, MSE loss, Adam (lr 3×10^{-3} , cosine decay), 40 epochs, batch 1024, the epoch with the lowest validation MSE on held-out trajectories is kept.

AR(1) stochastic closure (reimplemented in the spirit of [5]). g is the polynomial above and, with residual $r_k(t) = U_k(t) - g(X_k(t))$,

$$e_k^{n+1} = \phi e_k^n + \sigma \sqrt{1 - \phi^2} \xi_k^n, \quad \xi_k^n \sim \mathcal{N}(0, 1), \quad (4)$$

$$\phi = \frac{\langle r(t)r(t+\Delta) \rangle}{\langle r^2 \rangle}, \quad \sigma = \text{std}(r), \quad (5)$$

independent for every sector, started from its stationary distribution (the fast state, hence the true residual, is not available at forecast start).

Metrics. Forecast: from $N_{ic}=100$ truth initial conditions (10 on each of 10 test trajectories), an $M=10$ -member ensemble (identical members for deterministic closures) is run for 2 time units; `valid_time` is the first lead at which the ensemble-mean RMSE over sectors, divided by the standard deviation of X in a long reference run, exceeds 0.5 (mean over initial conditions; capped at 2); `rmse_0p5` is the normalised ensemble-mean RMSE at lead 0.5. Stability: 20 free runs of 400 time units from random reference states; `stable_frac` is the fraction that never has non-finite values or $|X| > 60$. Climate, pooled over the 8 statistically equivalent sectors and measured against a fixed long truth run: `mean_err` = $|\bar{X}_m - \bar{X}_t|$; `var_err` = $|\text{Var}_m / \text{Var}_t - 1|$; `w1_pdf` the 1-Wasserstein distance between the marginal distributions of X (computed on every seventh pooled sample of both runs); and `spec_err`, the RMS over $m=1..4$ of $\log_{10}(P_m^{\text{model}} / P_m^{\text{truth}})$, with P_m the time-mean power of the spatial Fourier mode m of X . Offline skill is $R^2 = 1 - \sum (U - \hat{g})^2 / \sum (U - \bar{U})^2$ on held-out trajectories.

IV. Experimental Setup

Tasks and data. Two regimes of the two-scale system: F20_c10 (primary; strong scale separation, $c=10$) and F20_c4 ($c=4$, weaker separation); $K=8$, $J=32$, $h=1$, $b=10$, $F=20$. The full model is integrated with RK4 at step 0.0025 (a logged sanity run, `experiments/precheck_dt.py`: at $c=10$, the mean and the variance of X over 40 trajectories and a 20-unit window each differed by less than 1% from a run with half the step from the same initial states) and sampled every 0.005. For seed s we generate independent trajectory sets after a 20-unit spin-up: 16 training chains of 25 time units (thinned by 4 for fitting), 4 validation chains (MLP early stopping only), 10 test chains of 30 units (forecast initial conditions every 3 units, offline R^2). Climate is scored against one fixed reference run (40 chains $\times$ 100 units, same for every system and seed). As a noise floor, an independent truth run of the same size (seed-dependent) is scored against that reference (“Truth (indep. run)”).

Protocol. Systems: no closure, polynomial (degree 4), MLP (1–32–32–1, local input), AR(1) stochastic (degree-4 polynomial plus noise); all reimplemented here, with no external code. Five seeds (0–4) for the main comparison and three (0–2) for ablations; a seed changes the training, validation and test trajectories, the MLP initialisation

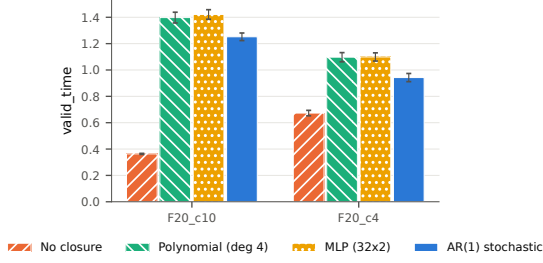


Fig. 1. Forecast valid time by system and task (mean over 5 seeds; error bars: standard deviation over seeds).

and batch order, and all noise streams. Rollouts: closed-model step 0.005 for every system, 10-member forecast ensembles, 20 free runs of 400 units with 10 units discarded, X stored every 0.05. Hyperparameters were fixed before any logged run and not tuned (no tuning budget used); the polynomial degree and the MLP input width are examined only as ablations on the test protocol (which has no separate selection split), so the ablation rows are not a model selection. Tests: rh compare reports Welch and paired t tests between per-seed values; we quote the Welch p without multiple-comparison correction, with $n=5$ per system.

Hardware and runtime. A shared macOS CPU machine, 2 threads, two experiment processes at a time; 99 runs in the registry for the study (including one smoke-test sanity run), plus the step-size pre-check logged afterwards as a second sanity run, with a median in-process time of about 10 s for non-neural systems and a median of about 44 s for MLP runs. Two MLP runs of the main group (seed 1 at $c=10$ and seed 4 at $c=4$) took 390 s and 549 s in-process because of contention on the shared machine, longer than the 6-minute limit we aimed for; their results are included. The registry also holds 24 superseded rows of the forcing-shift ablation: its 18 runs were first logged with shift-specific task labels, which the harness check does not accept, and the six AR(1) ones were logged a second time under an interim name; all 24 rows were superseded (rh supersede) and the 18 runs were executed again with the same commands and seeds under the final names. All metrics of the superseded rows other than the runtime are identical to those of the final rows. Code, per-run logs and the run registry (results/runs.jsonl, which holds the metrics of every run) are in the repository; the per-run JSON files and the generated data are not committed. experiments/run_all.sh lists the commands of the pre-check and of all main, floor and ablation runs under their final names.

V. Results

H1 (all learned closures beat no closure on valid time): supported. In Table I, at $c=10$ valid time rises from 0.363 (no closure) to 1.398 (polynomial), 1.421 (MLP) and 1.252 (AR(1)); at $c=4$ from 0.673 to 1.097, 1.099 and 0.942. All

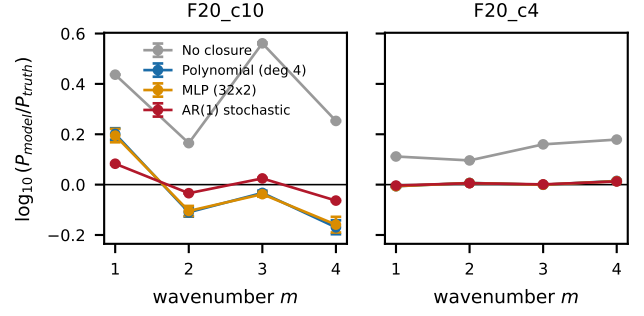


Fig. 2. Log ratio of the time-mean spatial power at wavenumber m to the reference (0 is perfect), mean $\pm$ std over 5 seeds.

six differences have Welch $p < 10^{-6}$ ($n=5$ each; rh compare with the no-closure reference). Normalised RMSE at lead 0.5 falls from 0.777 to 0.120–0.145 at $c=10$. This does not show skill beyond roughly one to one and a half time units: even the best closure has a mean valid time of 1.421, below the 2-unit forecast horizon.

H2 (AR(1) has a better climate than the deterministic polynomial): supported at $c=10$, not at $c=4$; with a cost in skill. At $c=10$ the deterministic closures give too little slow-variable variance (var_err 0.042 for the polynomial against 0.002 for AR(1), $p=0.0001695$) and a mis-shaped spectrum (spec_err 0.144 against 0.056, $p=0.0005618$); the AR(1) PDF distance is also lower (w1_pdf 0.083 against 0.243). Figure 2 shows the deterministic closures over-predict mode 1 and under-predict mode 4, which AR(1) reduces. AR(1) is still above the independent-truth floor in Table II for spec_err (0.056 against 0.013) and w1_pdf (0.083 against 0.027), but its var_err (0.002) equals the floor. At $c=4$ the picture is different: spec_err is 0.009 (polynomial), 0.008 (MLP), 0.007 (AR(1)), the difference polynomial–AR(1) has $p=0.3292$, and var_err is nominally worse for AR(1) (0.005 against 0.003, $p=0.1106$); all three are close to the floor (0.006 and 0.005), so the metrics cannot separate them. In both regimes AR(1) has a shorter valid time than the polynomial: 1.252 against 1.398 at $c=10$ ($p=0.0002893$) and 0.942 against 1.097 at $c=4$ ($p=8.141 \times 10^{-5}$; rh compare with the polynomial as reference). Figure 3 places the $c=10$ configurations run at the training forcing on this skill–climate plane.

H3 (an MLP with the same local input is not better than the polynomial): supported. Offline R^2 is 0.850 against 0.847 (gap below the registered 0.01 threshold, although a paired t test would detect it). At $c=10$ valid time is 1.421 against 1.398 ($p=0.3608$) and spec_err 0.137 against 0.144 ($p=0.6665$); at $c=4$ the differences are even smaller (1.099 against 1.097, $p=0.9246$). With 5 seeds this is absence of evidence of a difference rather than proof of equivalence; the paired test on valid time at $c=10$ is borderline ($p=0.05077$). A one-input function of X_k

TABLE I

Main comparison (mean $\pm$ std over 5 seeds). valid_time in model time units (higher is better), stable_frac and offline_r2 higher is better, other columns lower is better. Bold = best, underline = second best within a task.

Method	Task	valid_time $\uparrow$	rmse_0p5 $\downarrow$	stable_frac $\uparrow$	offline_r2 $\uparrow$	mean_err $\downarrow$	var_err $\downarrow$	w1_pdf $\downarrow$	spec_err $\downarrow$	n
No closure	F20_c10	0.363 $\pm$ 0.004	0.777 $\pm$ 0.019	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.439 $\pm$ 0.015	1.162 $\pm$ 0.008	1.895 $\pm$ 0.006	0.386 $\pm$ 0.002	5
Polynomial (deg 4)	F20_c10	1.398 $\pm$ 0.041	0.127 $\pm$ 0.007	1.000 $\pm$ 0.000	0.847 $\pm$ 0.001	0.236 $\pm$ 0.038	0.042 $\pm$ 0.007	0.243 $\pm$ 0.037	0.144 $\pm$ 0.020	5
MLP (32x2)	F20_c10	1.421 $\pm$ 0.036	0.120 $\pm$ 0.006	1.000 $\pm$ 0.000	0.850 $\pm$ 0.001	0.237 $\pm$ 0.046	<u>0.041 $\pm$ 0.008</u>	0.245 $\pm$ 0.045	<u>0.137 $\pm$ 0.023</u>	5
AR(1) stochastic	F20_c10	1.252 $\pm$ 0.029	0.145 $\pm$ 0.008	1.000 $\pm$ 0.000	0.847 $\pm$ 0.001	0.063 $\pm$ 0.007	0.002 $\pm$ 0.001	0.083 $\pm$ 0.003	0.056 $\pm$ 0.002	5
No closure	F20_c4	0.673 $\pm$ 0.021	0.378 $\pm$ 0.030	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.252 $\pm$ 0.011	0.341 $\pm$ 0.003	0.851 $\pm$ 0.002	0.141 $\pm$ 0.001	5
Polynomial (deg 4)	F20_c4	1.097 $\pm$ 0.035	0.158 $\pm$ 0.021	1.000 $\pm$ 0.000	0.719 $\pm$ 0.003	0.007 $\pm$ 0.005	0.003 $\pm$ 0.002	0.030 $\pm$ 0.007	0.009 $\pm$ 0.002	5
MLP (32x2)	F20_c4	1.099 $\pm$ 0.031	0.157 $\pm$ 0.021	1.000 $\pm$ 0.000	0.720 $\pm$ 0.003	0.009 $\pm$ 0.008	0.003 $\pm$ 0.002	0.032 $\pm$ 0.006	0.008 $\pm$ 0.003	5
AR(1) stochastic	F20_c4	0.942 $\pm$ 0.031	0.179 $\pm$ 0.018	1.000 $\pm$ 0.000	0.719 $\pm$ 0.003	0.009 $\pm$ 0.006	0.005 $\pm$ 0.002	0.029 $\pm$ 0.005	0.007 $\pm$ 0.002	5

TABLE II

Noise floor: an independent truth run scored against the reference (5 seeds).

Method	Task	mean_err $\downarrow$	var_err $\downarrow$	w1_pdf $\downarrow$	spec_err $\downarrow$	n
Truth (indep. run)	F20_c10	0.012 $\pm$ 0.004	0.002 $\pm$ 0.001	0.027 $\pm$ 0.005	0.013 $\pm$ 0.006	5
Truth (indep. run)	F20_c4	0.020 $\pm$ 0.007	0.005 $\pm$ 0.001	0.031 $\pm$ 0.010	0.006 $\pm$ 0.001	5

TABLE III

H5 and noise amplitude, $c=10$, seeds 0–2 (mean $\pm$ std). The deterministic polynomial and the fitted AR(1) closure are the main-group rows of Table I (five seeds).

Closure noise	valid_time	var_ratio	w1_pdf	spec_err
AR(1) $\sigma \times 0.5$	1.367 $\pm$ 0.016	0.992 $\pm$ 0.002	0.078 $\pm$ 0.006	0.053 $\pm$ 0.002
AR(1) $\sigma \times 1.5$	1.129 $\pm$ 0.014	1.004 $\pm$ 0.002	0.104 $\pm$ 0.005	0.071 $\pm$ 0.003
White noise ($\phi=0$)	1.399 $\pm$ 0.025	0.985 $\pm$ 0.002	0.101 $\pm$ 0.009	0.067 $\pm$ 0.005

may simply be a smooth curve that a degree-4 polynomial already fits; we did not isolate this.

H4 (stability, and MLP degradation under forcing shift): stability supported but uninformative; the shift part is refuted. stable_frac is 1.000 for every system, task and seed (Table I) and in all ablation runs, so no closure diverged in 20 runs of 400 time units each; this does not exclude instabilities in other settings, in longer rollouts or with different inputs. Instabilities of offline-trained parametrisations have been reported for Earth system models [2]; we saw none in this low-dimensional system, which says nothing about those models. Under a forcing shift (trained at $F=20$, deployed at $F=18$ and 22; Table VI) the MLP’s w1_pdf is lower than the polynomial’s in both shifts (0.057 against 0.070 at 18, 0.034 against 0.039 at 22), the opposite of the hypothesis. The shift was mild: the deterministic closures’ climates at $F=18$ and 22 are closer to truth than at the training value $F=20$ (e.g., polynomial w1_pdf 0.070, 0.039 against 0.2427 in the main group), so $F=20$ appears to be the hardest of the three regimes for local closures and this experiment does not test extrapolation properly.

VI. Ablations and Analysis

H5 and noise amplitude (Table III). The ablation rows use seeds 0–2 and are set against the five-seed main-group values of Table I, so these comparisons are not seed-matched. White noise of the same variance ($\phi=0$) gives spec_err 0.067 against 0.05629 for the fitted AR(1)

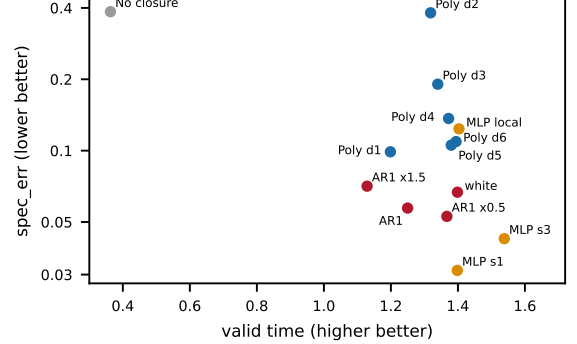


Fig. 3. Skill against climate at $c=10$ for every configuration run at the training forcing $F=20$: main systems and the noise, MLP-input and degree ablations (mean over seeds 0–2 for each point; log axis for spec_err). Colours: grey no closure, blue polynomial, orange MLP, red AR(1) variants.

TABLE IV

Polynomial degree, $c=10$, seeds 0–2. Degree 4 is the main-group polynomial of Table I (five seeds).

Polynomial	offline_r2	valid_time	mean_err	w1_pdf	spec_err
Degree 1	0.791 $\pm$ 0.001	1.199 $\pm$ 0.045	0.192 $\pm$ 0.007	0.197 $\pm$ 0.006	0.099 $\pm$ 0.005
Degree 2	0.830 $\pm$ 0.001	1.319 $\pm$ 0.017	0.598 $\pm$ 0.029	0.607 $\pm$ 0.029	0.381 $\pm$ 0.026
Degree 3	0.841 $\pm$ 0.001	1.340 $\pm$ 0.032	0.315 $\pm$ 0.094	0.321 $\pm$ 0.094	0.191 $\pm$ 0.057
Degree 5	0.847 $\pm$ 0.001	1.380 $\pm$ 0.032	0.167 $\pm$ 0.016	0.174 $\pm$ 0.016	0.105 $\pm$ 0.006
Degree 6	0.848 $\pm$ 0.001	1.394 $\pm$ 0.036	0.179 $\pm$ 0.033	0.188 $\pm$ 0.033	0.109 $\pm$ 0.018

(ϕ fitted to 0.982 for seed 0), and both are far below the deterministic closure (0.1435); white noise retains a longer valid time (1.399 against 1.252). The ordering in H5 holds in the means, but the gap is small compared with the gap to the deterministic closure, so in these means most of the reduction in spec_err is already obtained with white noise of the fitted variance, and we cannot attribute the remainder to temporal correlation without more seeds or a test; the white-noise run also has a lower variance ratio (0.985 against 0.9979). Amplitude trades skill for climate: valid time falls monotonically from 1.398 (no noise) through 1.367 ($\times 0.5$) and 1.252 to 1.129 ($\times 1.5$), while spec_err is lowest at $\times 0.5$ (0.053) and similar at $\times 1$ (0.05629) and worse at $\times 1.5$ (0.071). The fitted amplitude is therefore not obviously optimal: half of it gave a similar climate and a valid time close to that of the deterministic closure (3 seeds against 5, no test).

TABLE V

MLP input width, $c=10$, seeds 0–2 (hidden width 32 in all rows). The local-input MLP is the main-group row of Table I (five seeds).

MLP input	offline_r2	valid_time	mean_err	w1_pdf	spec_err
X_{k-1}, X_k, X_{k+1}	0.878 ± 0.002	1.398 ± 0.013	0.019 ± 0.003	0.056 ± 0.004	0.031 ± 0.004
$X_{k-3..k+3}$	0.923 ± 0.001	1.538 ± 0.012	0.053 ± 0.049	0.062 ± 0.043	0.043 ± 0.024

Polynomial degree (Table IV), a failure case. Over the ablation degrees 1, 2, 3, 5 and 6, offline R^2 never decreases with degree (0.791 to 0.848), and valid time increases with it (1.199 to 1.394), but the climate does not: degree 2 has by far the worst climate (spec_err 0.381, mean_err 0.598), worse than the linear closure (0.099), while degree 3 is at 0.191, degrees 5 and 6 are at 0.105 and 0.109, and degree 4 (main group, five seeds) is at 0.1435. We have no isolating experiment for why degree 2 fails and do not claim a mechanism; the point is that offline R^2 and forecast skill did not predict the climate of the coupled model.

MLP input width (Table V). Giving the MLP the neighbouring slow variables lifts offline R^2 from 0.8497 (local input, main group, five seeds) to 0.878 (three inputs) and 0.923 (seven inputs). Climate improves at width three (spec_err 0.1375 to 0.031, mean_err 0.2372 to 0.019) while valid time does not (1.398 against 1.421); the seven-input net had the best valid time (1.538) but a large seed-to-seed spread of mean_err (0.053 ± 0.049). This exploratory result (3 seeds, not seed-matched to the main group) suggests that the local-input restriction, rather than the function class, may be what limited the MLP in the main comparison; it was not a registered hypothesis and the width was not chosen on a separate split.

Forcing shift (Table VI). No closure diverged. At $F=18$ the AR(1) closure is worse than the deterministic ones on spec_err (0.078 against 0.039 for the polynomial) and w1_pdf (0.107 against 0.070), whereas at $F=20$ it was better; its noise amplitude was fitted at $F=20$ and kept fixed, which may explain it (not isolated by a run). At $F=22$ AR(1) is again behind the polynomial on spec_err (0.015 against 0.010) and w1_pdf (0.046 against 0.039) and has the largest var_err (0.017), while the MLP is best on all climate columns of the table.

VII. Limitations

Two-scale Lorenz-96 is an idealised toy system, and the ranking of closures found here may not transfer to real atmosphere or ocean models. This is a small CPU study of one 8-variable configuration ($K=8$, $J=32$, $F=20$) with two values of c , five seeds in the main comparison and three in ablations; the p -values are uncorrected Welch tests on $n \leq 5$ and the equivalence claim in H3 rests on non-significance. All baselines are our reimplementations with fixed, untuned hyperparameters; a tuned MLP or a different closure family could change the ranking. All closures depend only on the local slow variable in the main comparison and are trained offline: we do not test coupled/online training, which the cited work proposes as a

remedy for instabilities and biases [2], and “stable” means stable over 20 runs of 400 time units. The reference climate is itself a finite run, and the independent-truth floor shows how large that sampling error is (Table II). The AR(1) noise is independent between sectors and uninitialised at forecast start, so its forecast skill reflects an ensemble mean with 10 members. The ablations reuse the test protocol and the forcing shift is mild and one-dimensional; the degree-2 failure and the noise-amplitude effects are not explained mechanistically. Two runs exceeded the intended 6-minute wall time (Section IV). The step-size pre-check is a single run that compares only the mean and variance of X . A full study would add more seeds, tuned neural models, state-dependent and multiplicative noise, online training, ensemble-skill scores (CRPS) and larger systems.

VIII. Conclusion

In this small two-scale Lorenz-96 study, all three learned closures extended the forecast valid time well beyond no closure and none diverged. The deterministic polynomial and MLP with a local input were not distinguished by the Welch tests on valid time and spectral error at 5 seeds, the AR(1) closure had the best climate at $c=10$ but a shorter valid time, and at $c=4$ the climate metrics did not separate the learned closures from the noise floor. In the three-seed ablations, a better offline fit and forecast skill did not come with a better climate (degree 2 against degree 1), the AR(1) closure lost its climate advantage at the two shifted forcings (its noise amplitude, fitted at $F=20$, may not transfer), and giving the MLP neighbouring inputs may matter more than the choice of function class; these are exploratory results in an idealised toy system.

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TABLE VI

Forcing shift: closures trained at $F=20$, slow model and truth at the deployment F ($c=10$, seeds 0-2). The values at the training forcing $F=20$ are in Table I (five seeds).

Deploy F	System	valid_time	mean_err	var_err	w1_pdf	spec_err
18	Polynomial (deg 4)	1.485 ± 0.043	0.014 ± 0.004	0.003 ± 0.001	0.070 ± 0.003	0.039 ± 0.003
18	MLP (32x2)	1.538 ± 0.050	0.019 ± 0.002	0.002 ± 0.000	0.057 ± 0.001	0.038 ± 0.000
18	AR(1) stochastic	1.367 ± 0.031	0.089 ± 0.003	0.004 ± 0.001	0.107 ± 0.006	0.078 ± 0.006
22	Polynomial (deg 4)	1.357 ± 0.044	0.036 ± 0.002	0.010 ± 0.001	0.039 ± 0.002	0.010 ± 0.001
22	MLP (32x2)	1.394 ± 0.037	0.003 ± 0.004	0.004 ± 0.001	0.034 ± 0.004	0.006 ± 0.001
22	AR(1) stochastic	1.211 ± 0.062	0.033 ± 0.002	0.017 ± 0.001	0.046 ± 0.002	0.015 ± 0.000

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